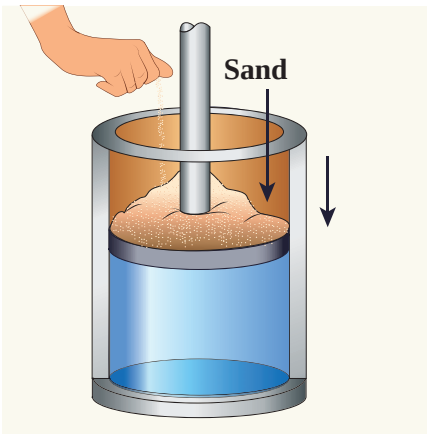
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A quasi-static process is an infinitely slow process in which the system changes its variables (P,V,T) so slowly such that it remains in thermal, mechanical and chemical equilibrium with its surroundings throughout. By this infinite slow variation, the system is always almost close to equilibrium state.

EXAMPLE 8.14

Give an example of a quasi-static process.

Consider a container of gas with volume V, pressure P and temperature T. If we add sand particles one by one slowly on the top of the piston, the piston will move inward very slowly. This can be taken as almost a quasi-static process. It is shown in the figure



Sand particles added slowly- quasi-static process

8.6.4 Work done in volume changes

Consider a gas contained in the cylinder fitted with a movable piston. Suppose the gas is expanded quasi-statically by pushing the piston by a small distance dx as shown in Figure 8.21. Since the expansion occurs quasi-statically the pressure, temperature and internal energy will have unique values at every instant.

The small work done by the gas on the piston

$$dW = Fdx \qquad (8.14)$$

The force exerted by the gas on the piston F = PA. Here A is area of the piston and P is pressure exerted by the gas on the piston.

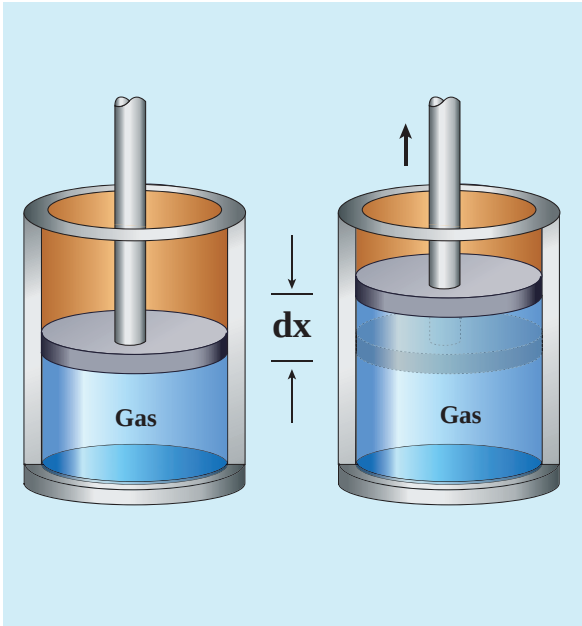


Figure 8.21 Work done by the gas

Equation (8.14) can be rewritten as

$$dW = PA\,dx \qquad (8.15)$$

But Adx = dV= change in volume during this expansion process.

So the small work done by the gas during the expansion is given by

$$dW = PdV \qquad (8.16)$$

Note here that is positive since the volume is increased. Here, is positive.

In general the work done by the gas by increasing the volume from V_i to V_f is given by

$$W = \int_{V_i}^{V_f} PdV \qquad (8.17)$$